MIL-I-60892(MU)
27 November 1968
SUPERSEDING
FA-PD-MI-2486, Rev. O
17 March 1967

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED: DELAY, M43A1
ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M43A1 Assembly, loaded with Cartridge, Delay, M113 for use in conjunction with other cartridge actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167	Ammunition Data Card
MS28741	Hose Assembly - Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command	
F10520224	- Initiator, Cartridge Actuated, Delay, M43, Assembly
C8798143	- Container, Ammunition, Fiber M375 for Initiators, Delay, M4, M6A1, M12A1, M26, M33, M45A1 or M49A1
C8798142	- Box, Packing, Ammunition for Initiators, Delay, M4, M6A1, M12A1, M26, M33, M45A1 or M49A1, in Fiber Container, M375
IEL-10520224	- Inspection Equipment List for Initiator, Cartridge Actuated, Delay, M43, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin, cartridge, cartridge retainer and chamber shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inches, maximum.

3.3 Initiator pin security.- The initiator pin of the assembled initiator, less cartridge, cartridge retainer and chamber shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall cause no deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

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3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, size No. 4, as specified in MS28741.

TABLE I

Temperature	Pressure (PSI)	Ignition Delay (See 6.3)
-65° to -70°F	500 to 1500	4.2 Seconds, Max.
65° to 75°F	575 to 1500	3.0 ± .30 Seconds
160° to 165°F	625 to 1500	2.30 Seconds, Min.

3.6.2 The initiator shall not fail to fire.

3.6.3 The initiator shall function without mechanical failure (See 6.2).

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawing, referenced specifications and the following:

3.7.1 Metal defects.- The component shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred percent examinations shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Delay, M113, Drawing D8596114 covering a detail of Drawing F10520224.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" ring missing	Visual
4. Sealing disc missing or damaged	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. "O" ring split, cut or cracked	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing F10520224, less Drawing D8596114, D8594860, D8594857 and B8359536.

CRITICAL	
1. Mixed initiators	Visual
2. Initiator not in cocked position	Visual
3. Safety pin missing or improperly engaged	Visual
MAJOR	
101. "O" ring missing or not lubricated	Visual
102. Thread damaged	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly, Drawing F10520224, less Drawings D8594857 and B8359536.

CRITICAL	
1. Cartridge loose	Visual and Manual
2. Cartridge retainer loose	Visual and Manual
3. Metering groove of cartridge retainer missing	Visual
4. Elongated hole of cartridge retainer missing	Visual

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Drawing F10520224, less Drawing B8359536.

CRITICAL	
1. Cartridge retainer improperly positioned	Gage
2. Any part missing	Visual
3. Improper assembly	Visual
4. Foreign matter in gas port	Visual
MAJOR	
101. Length from cap boss face to gas port face, max.	Gage
102. Protrusion of initiator pin, max.	Gage
103. Initiator pin not free to rotate	Manual
104. Safety pin loose or deformed	Manual and Visual
105. Thread damaged or not full	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Drawing F10520224.

CRITICAL	
1. Any part missing, except as otherwise classified	Visual
2. Marking missing, incorrect or illegible	Visual

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4.2.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Height of cap, max.	Gage
102. Set screw missing, loose, above flush or not covered with barrier material	Visual
MINOR	
201. Total length, max.	Gage
202. Location of labels incorrect	Visual
203. Shipping cap missing	Visual
204. Protective finish inadequate or defective	Visual

4.2.2.6 Fiber Container, Drawing C8798143.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Packing Box, Drawing C8798142.

CRITICAL	
None defined	
MAJOR	
None defined	

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4.2.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual
203. Strapping missing, loose, broken or improperly assembled	Visual and Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Firing pin release.- The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.- Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the assembly.

4.2.3.5 Functional.- The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics.- Twenty one units shall be tested to determine compliance with the requirements of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in Table III and 4.3.5.1.1. The variables sampling plan of MIL-STD-414, example C-2 or C-3 as applicable for the AQL listed in Table III shall be used for each sample of 7 units. Failure of any unit to comply with the requirements of 3.6.1, 3.6.2 and 3.6.3 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

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4.2.3.5.1 (Cont'd)

TABLE III

Temperature	AQL	Sample Size
-65° to -70°F	0.65%	7
65° to 75°F	0.65%	7
160° to 165°F	0.65%	7

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10520224.

4.3 Test methods and procedures.

4.3.1 Firing pin release.- The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.3.2 Initiator pin security.- The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.3.3 Breakaway torque.- A torque wrench, set at the torque requirement specified in 3.4 shall be used in conjunction with a fixture to determine the adequacy of the joint. The torque shall be applied to the cap.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The samples shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperature.

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4.3.5.1.1 Temperature conditioning.- A standard M43A1 initiator, loaded with a dummy M113 cartridge containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8798143 and C8798142 except that all markings shall refer to Initiator, Cartridge Actuated, Delay, M43A1.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement document should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

6.4 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2486, Rev. 0 dated 17 March 1967.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A253

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Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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